

Can the Vertical Shear Instability Survive Dust Coagulation and Settling?

Simulations with the **TriPoD** Dust Coagulation Model

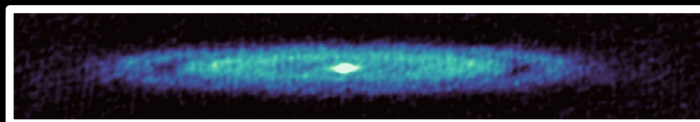
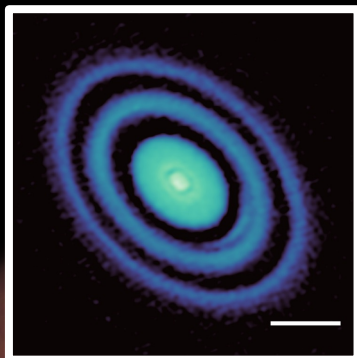
Thomas Pfeil

Collaborators: Phil Armitage, Til Birnstiel, Yan-Fei Jiang, Linn Eriksson,
Nicolas Kaufmann, Hubert Klahr, Vignesh Vaikundamaran, Alexandros Ziampras

Vertical Shear Instability

*Urpin and Brandenburg (1998), Nelson+ (2013), Lin and Youdin (2015),
and many more*

Thickness of the dust layer (observations)



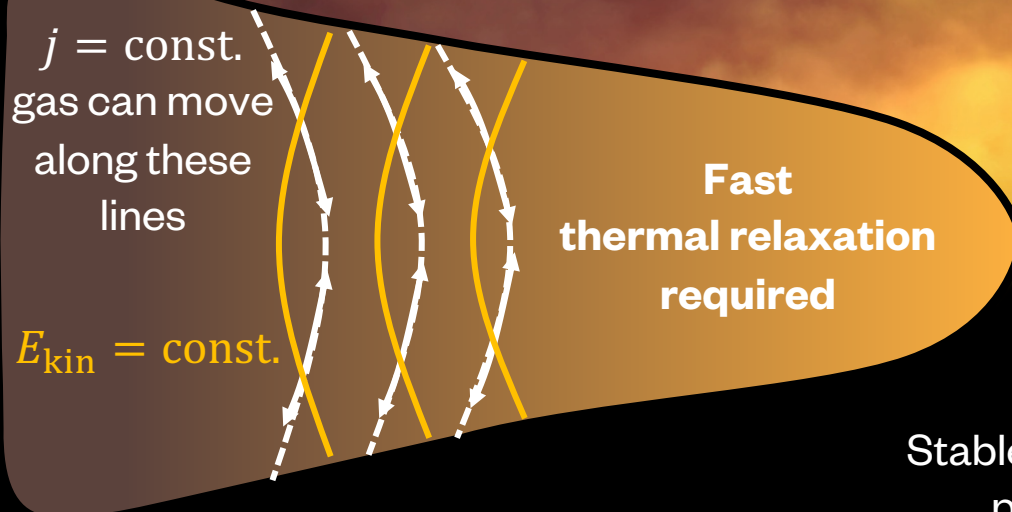
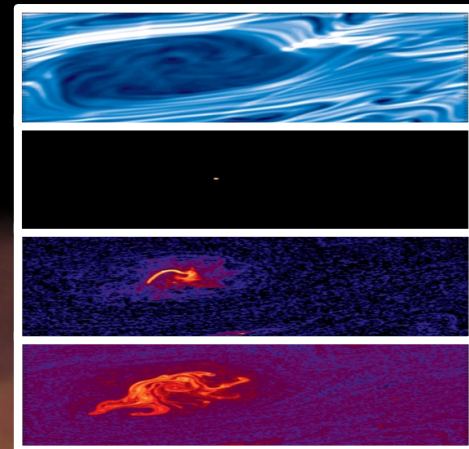
Oph 163131

(*Villenave et al., 2022*)

HD 163296

(*Andrews et al., 2018*)

Dust trapping vortices
→ planetesimal formation
e.g., *Raettig et al. 2021*

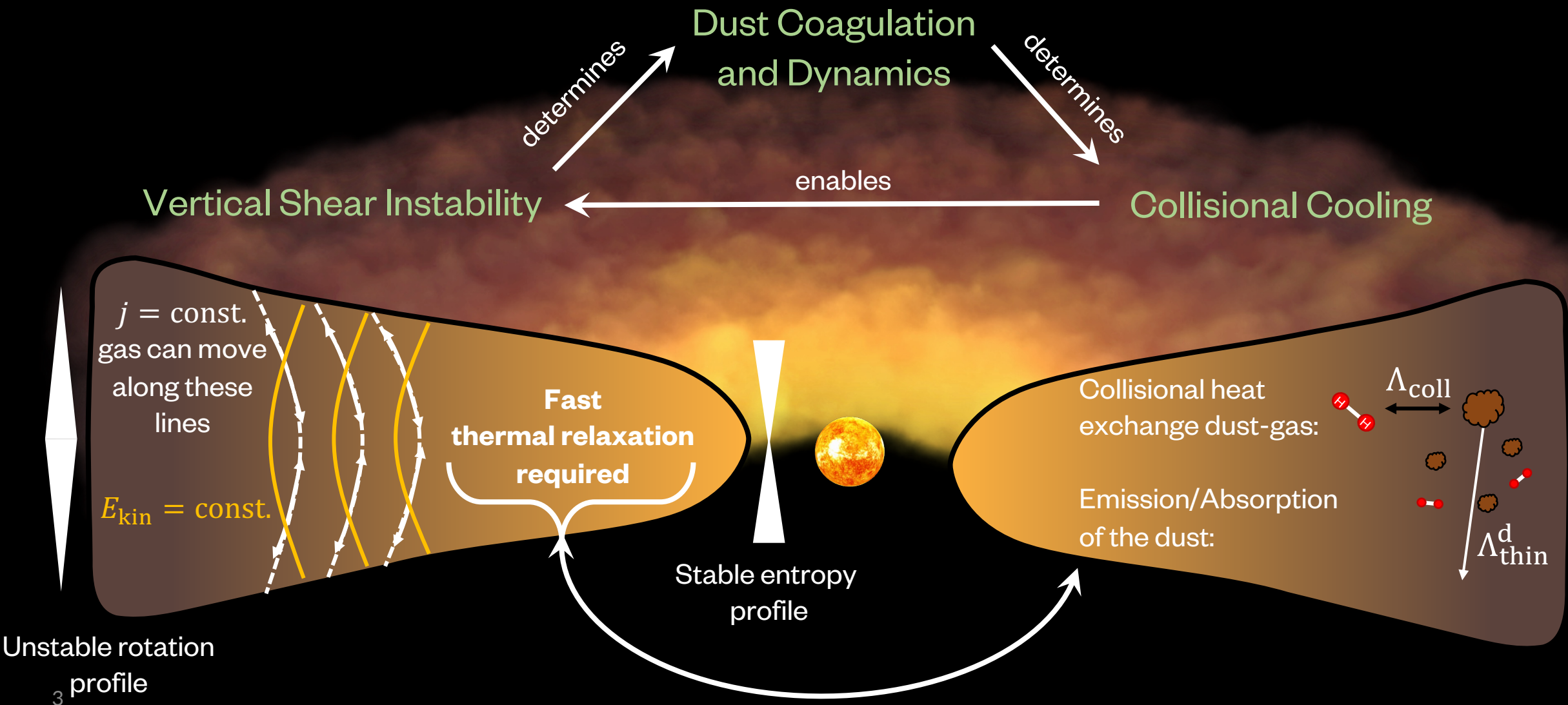


Stable entropy
profile

Unstable rotation
2 profile

Vertical Shear Instability

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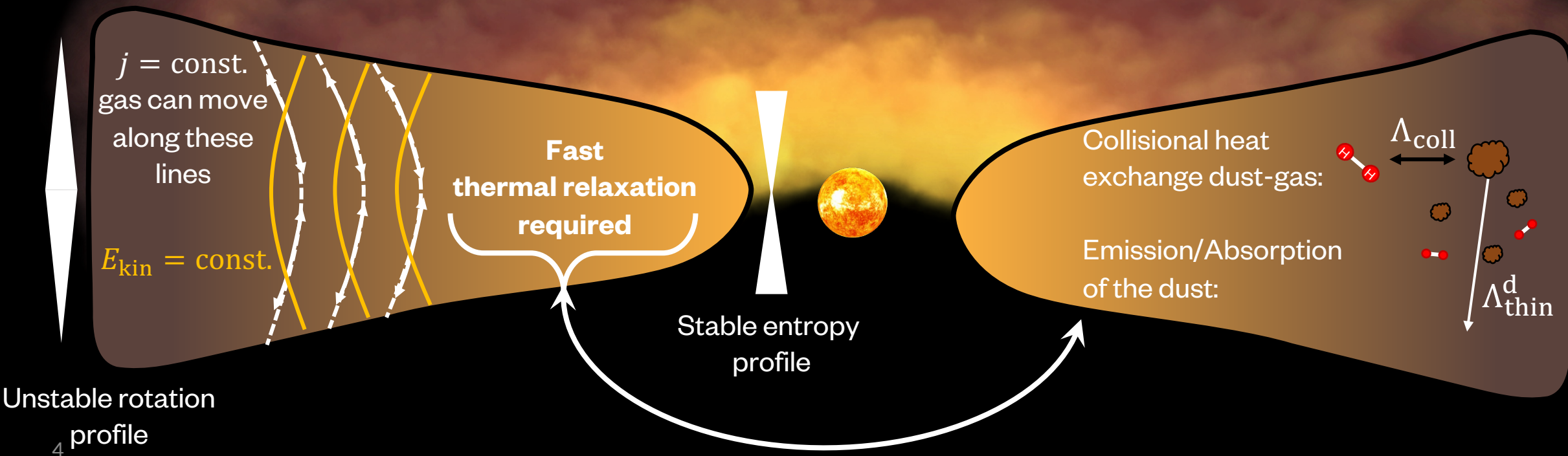


Vertical Shear Instability

A more realistic cooling function

$$t_g^{\text{coll}} = \frac{\gamma}{\gamma - 1} \frac{1}{n_S \sigma_S \bar{v}_g} \quad \text{Coagulation}$$

$$t_d^{\text{rad}} \approx \frac{C_d}{16 \sigma_{\text{SB}} T_{\text{eq}}^3 \langle \kappa_P(a_{\text{min}}, a_{\text{max}}, q, T) \rangle \rho_d}$$

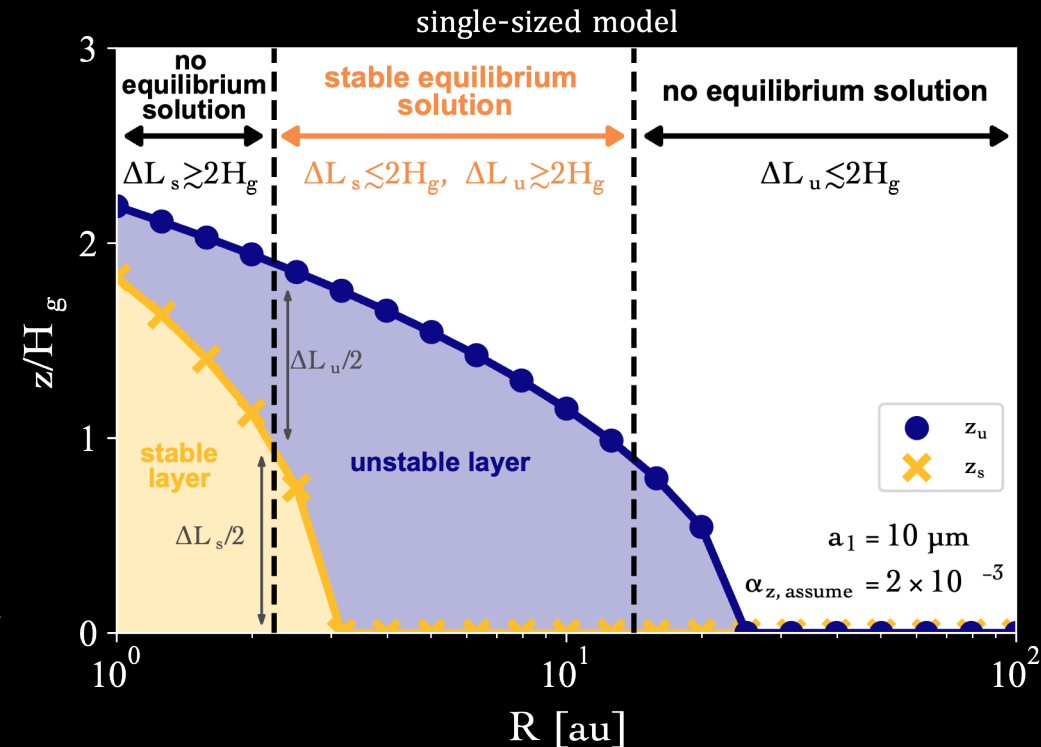
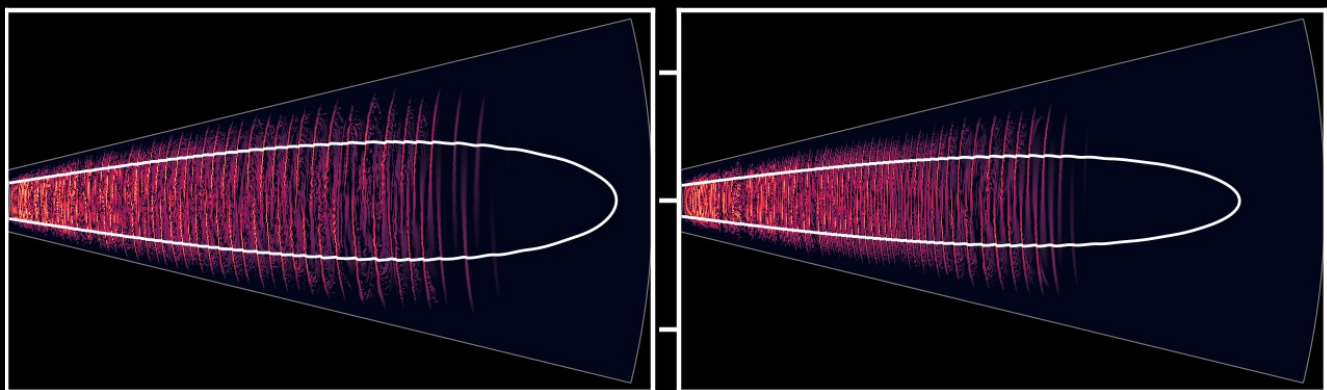


Vertical Shear Instability

Incorporating a realistic cooling function

- Dust coagulation determines the cooling times of the gas via dust-gas collisions (Fukuhara+ 2021, 2024; Pfeil+ 2023, 2024)
 - larger grains
 - inefficient collisional cooling
- 2D $r - \vartheta$ simulations required

Increasing fragmentation velocity (larger particles)



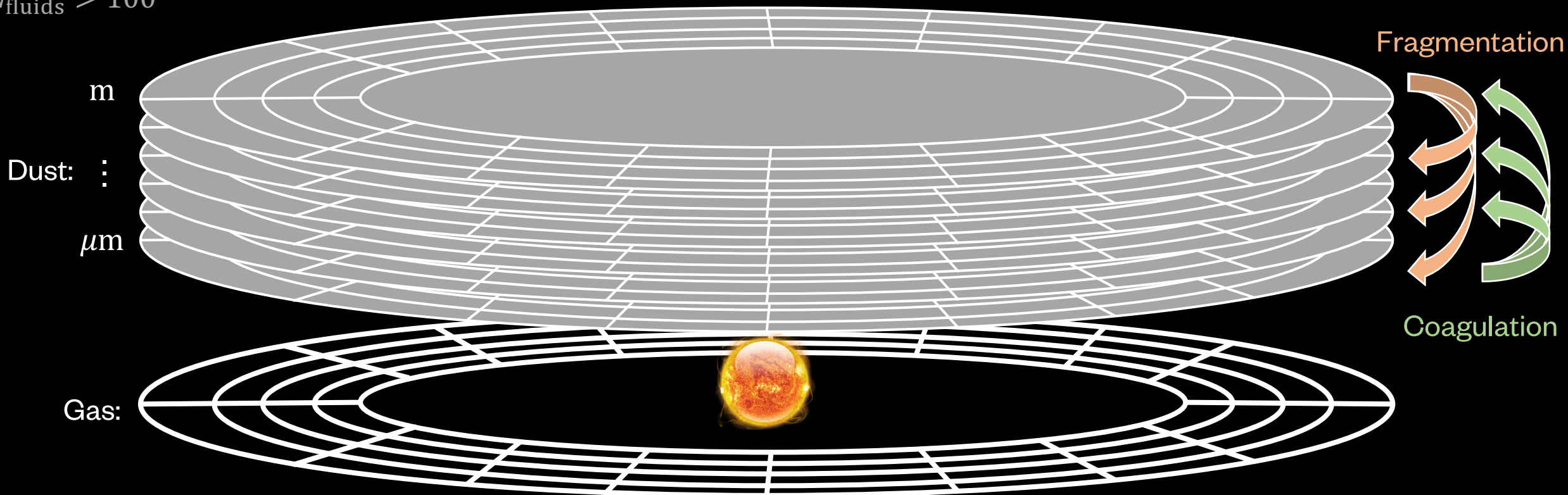
Adapted from *Fukuhara+ (2024)*
VSI stable zones based on a dust model with single-sized 10 micron particles

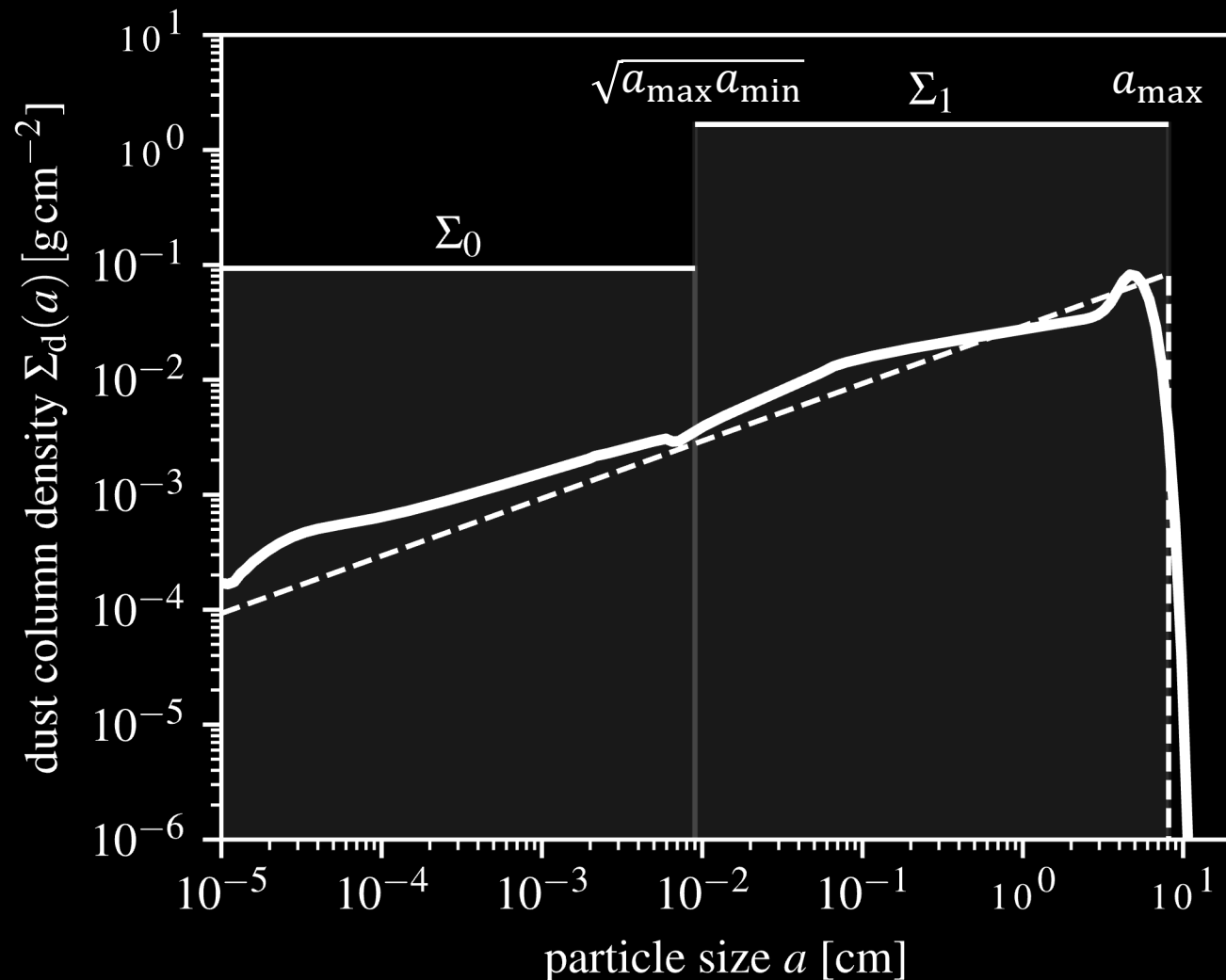
$$N_{\text{cells}} = N_r \times N_\phi \times N_{\text{fluids}}$$

$$N_{\text{fluids}} > 100$$

Smoluchowski coagulation equation
as in *Stammler+ (2022)*:

$$\frac{\partial n_k}{\partial t} = \sum_{i=1}^{N_m} \sum_{j=1}^i K_{ijk} R_{ij} n_i n_j - n_k \sum_{j=1}^{N_m} n_j R_{jk} (1 + \delta_{jk})$$





We define two bins,
a maximum, and a minimum particle size:

$$\Sigma_0 = \int_{a_{\min}}^{\sqrt{a_{\max} a_{\min}}} \Sigma_d(a) da$$

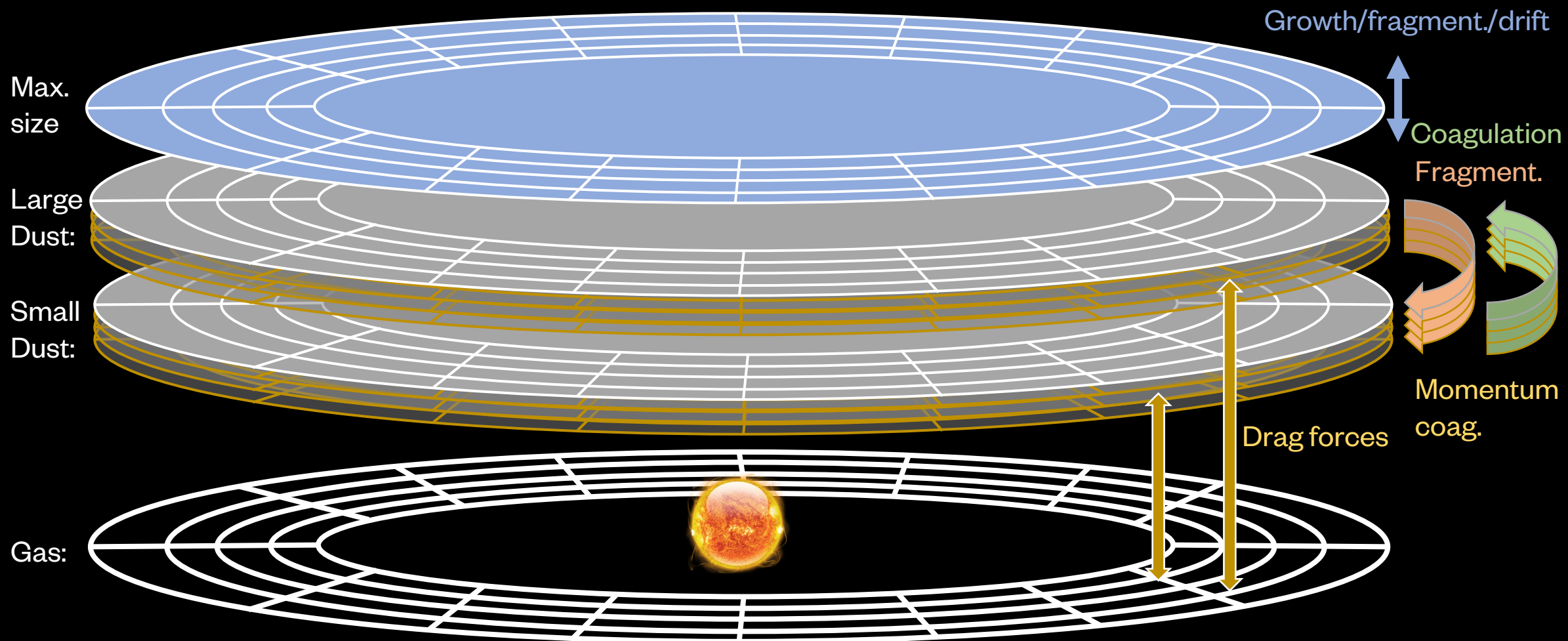
$$\Sigma_1 = \int_{\sqrt{a_{\max} a_{\min}}}^{a_{\max}} \Sigma_d(a) da$$

With this, we can describe the size
distribution as a power law $\Sigma_d(a) \propto a^{q+4}$

$$q = \frac{\log(\Sigma_1/\Sigma_0)}{\log(a_{\max}/\sqrt{a_{\max} a_{\min}})} - 4$$

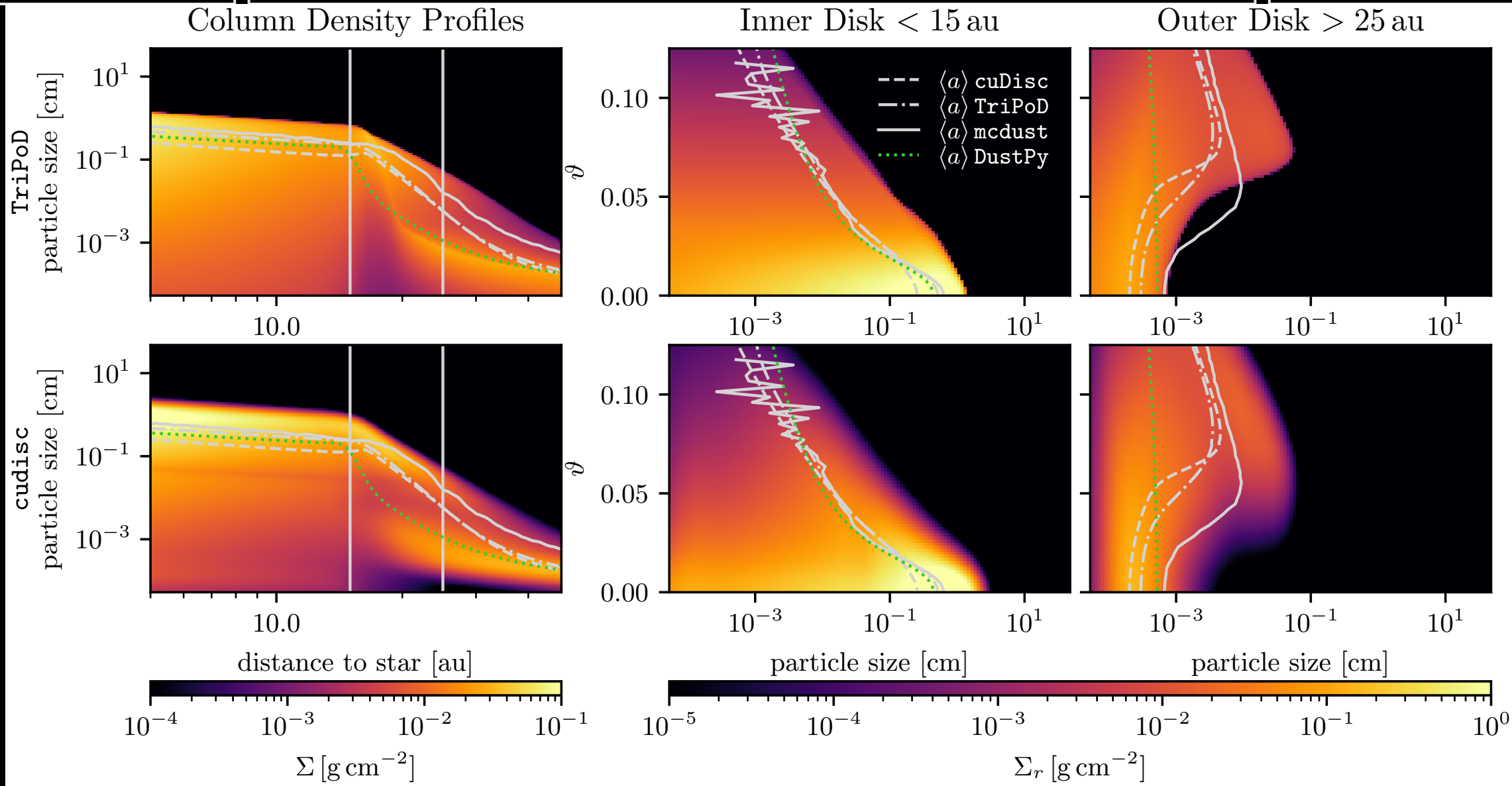
Dust Evolution: Coagulation

Numerical Methods: TriPoD (*Pfeil+ 2024*)



Dust Evolution: Coagulation

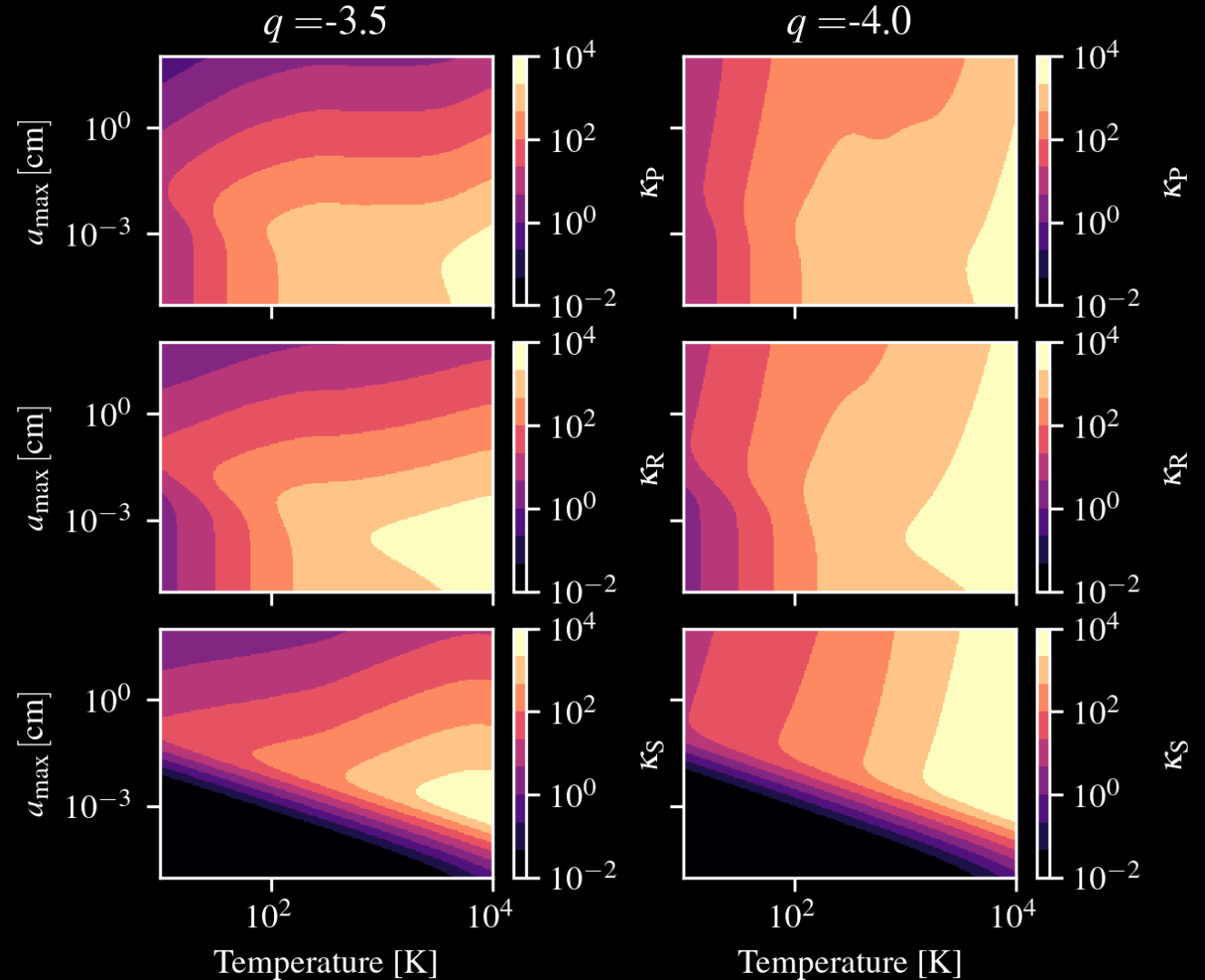
Code Comparison (Eriksson, Pfeil, Kaufmann, Vaikundamaran, 2026)



- Dust opacity tables from `dsharp_opac` (Birnstiel+, 2018) for a grid of maximum particle sizes, power law indices and dust temperatures
- We interpolate these 3D tables to the given power-law size distribution `TriPoD` provides:

$$n(a) \propto \begin{cases} a^q & \text{if } a < a_{\max} \\ 0 & \text{if } a > a_{\max} \end{cases}$$

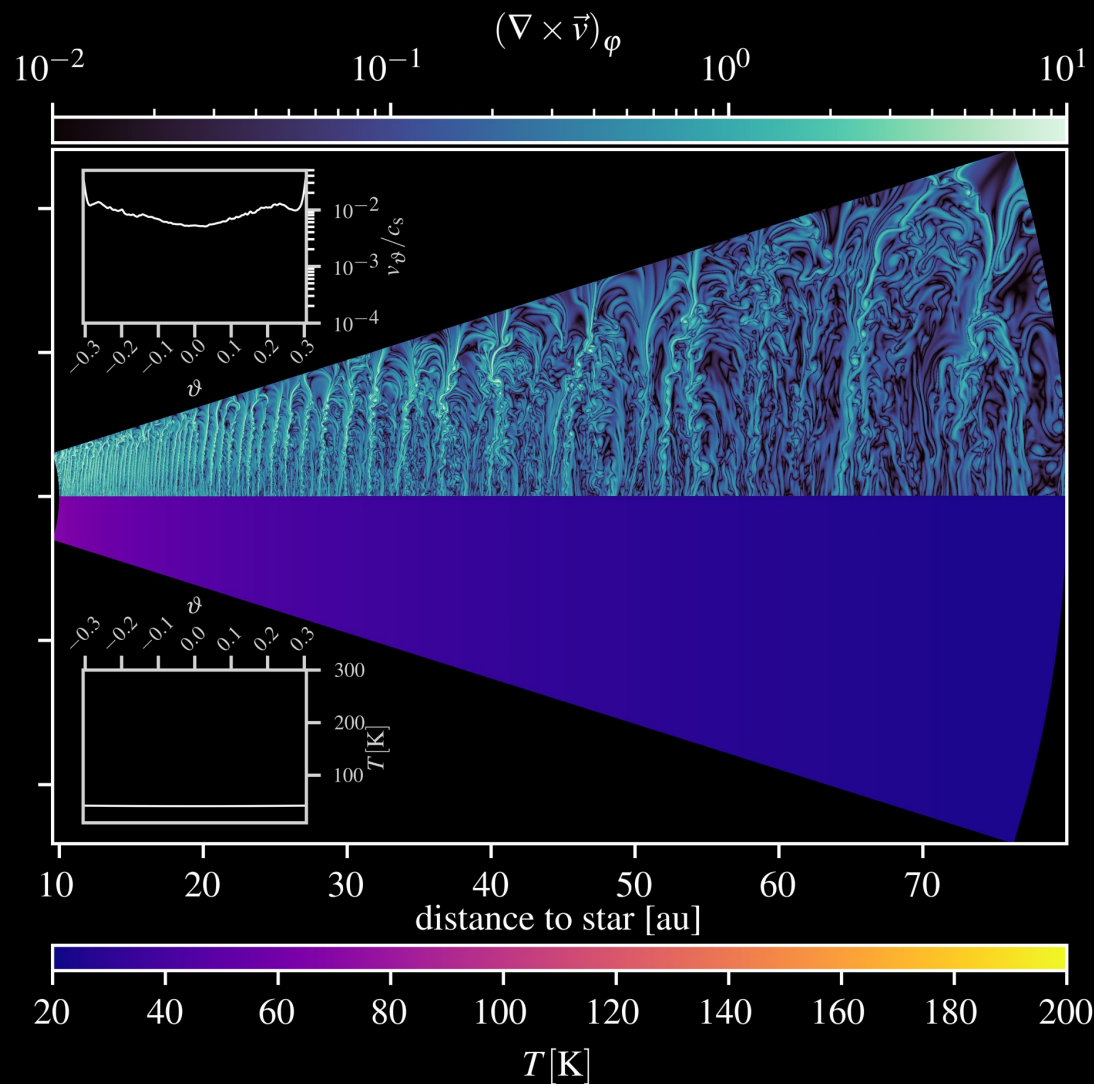
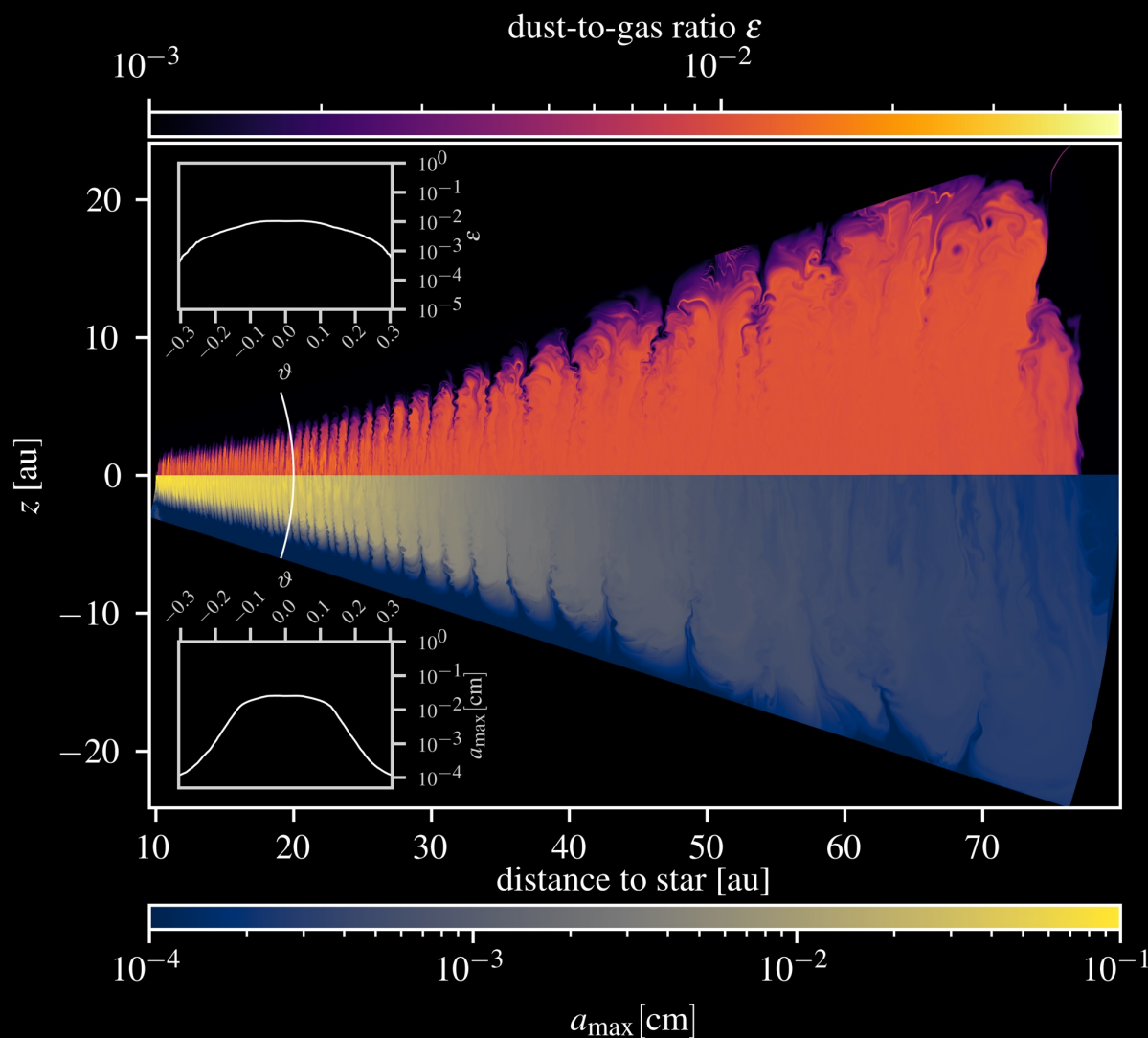
See also `growcapacity`, by Alex
based on `optool`:



Vertical Shear Instability

Exploring the VSI's operational range

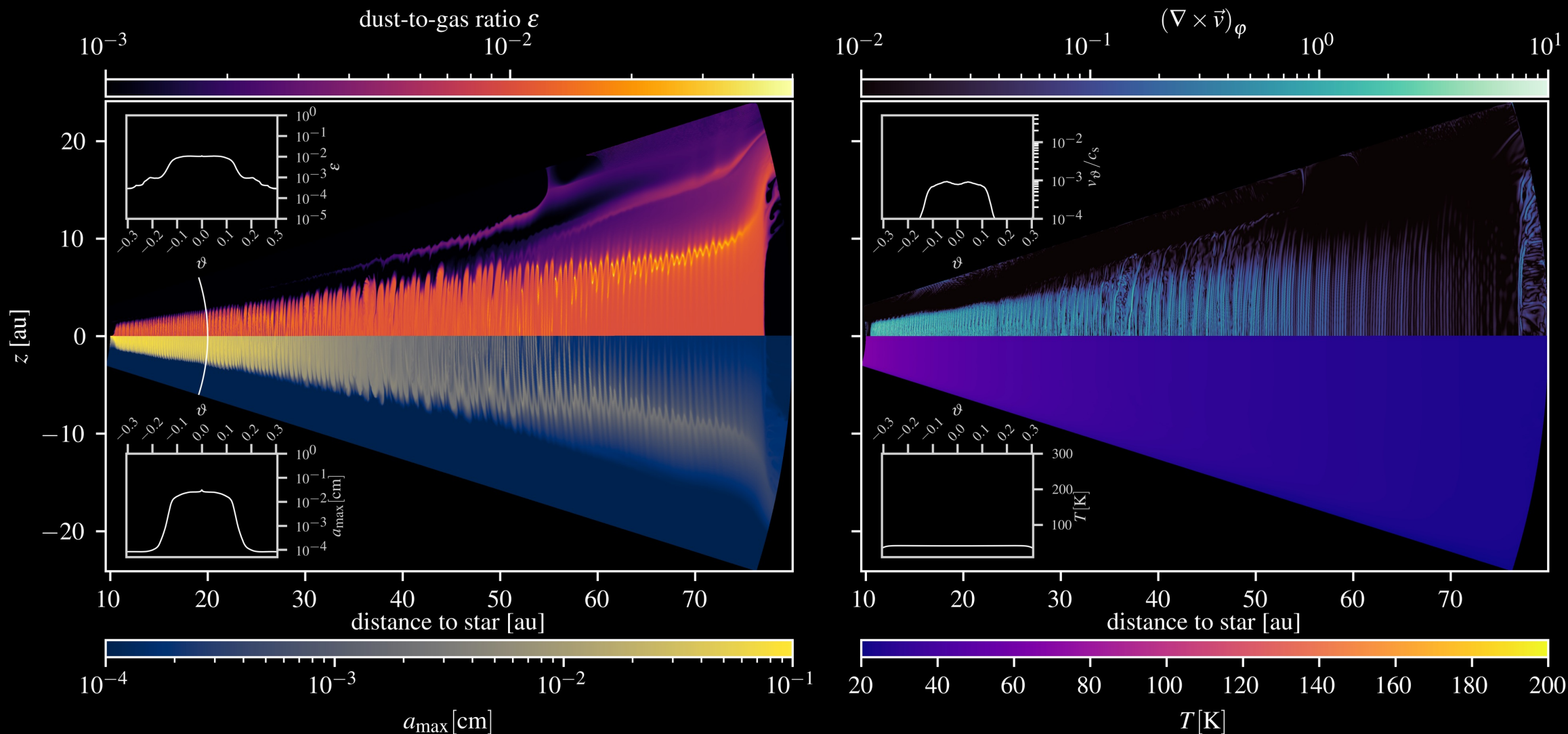
“**Ideal VSI**”: Isothermal simulation, no vertical temperature stratification, no dust feedback (dust is a passive scalar)



Vertical Shear Instability

Exploring the VSI's operational range

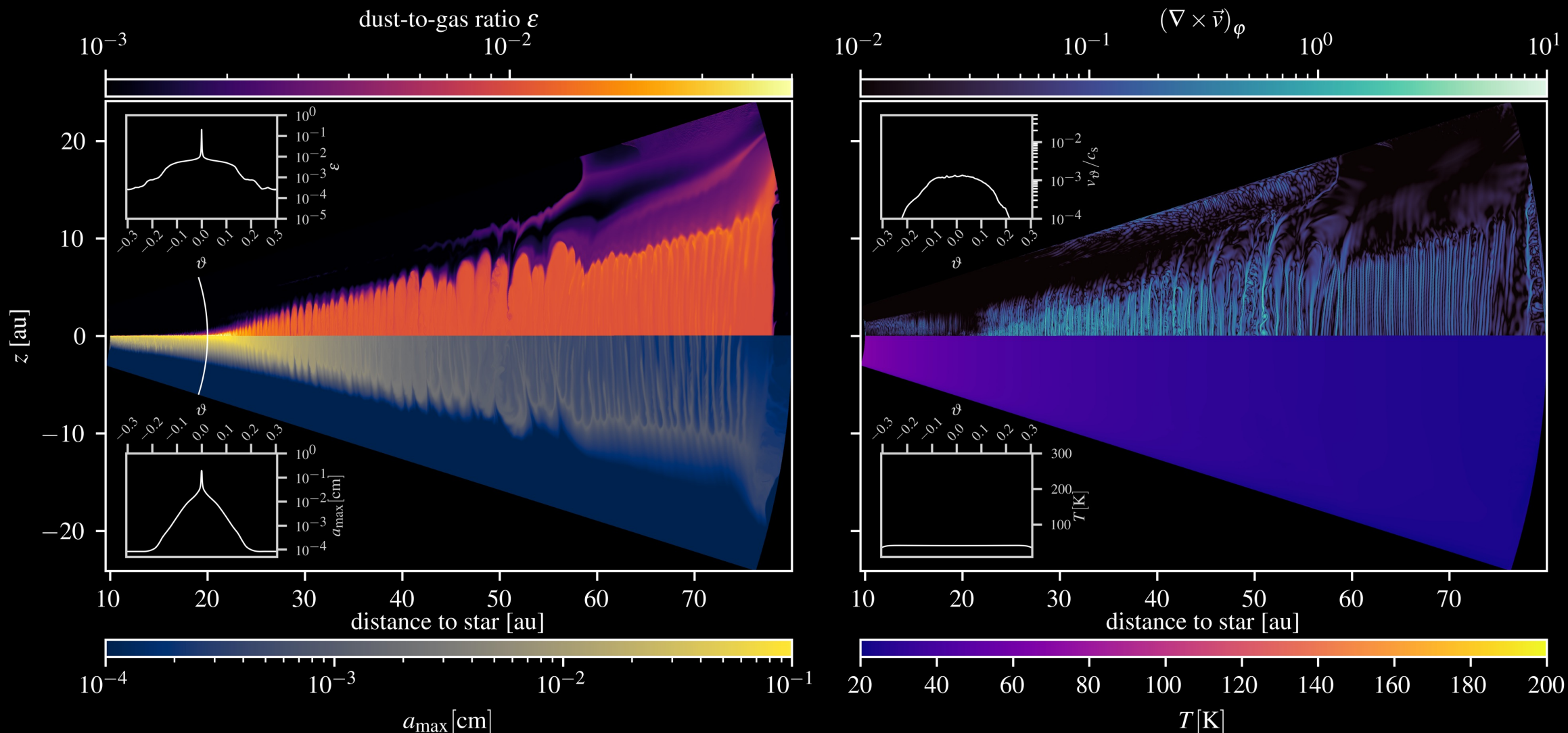
Dust-Gas cooling (non-isothermal), no vertical temperature stratification, no dust feedback (dust is a passive scalar)



Vertical Shear Instability

Exploring the VSI's operational range

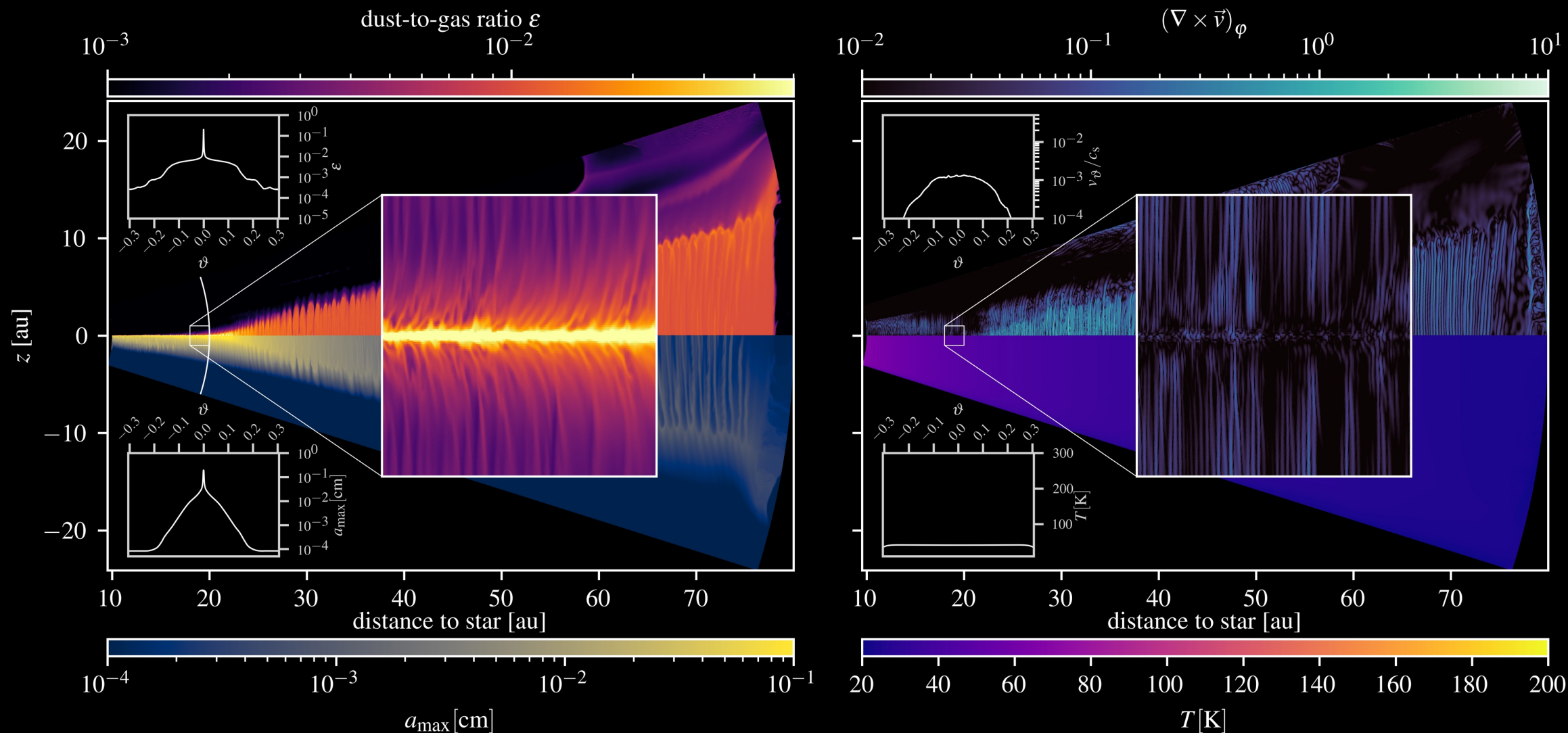
Dust-Gas cooling (non-isothermal), no vertical temperature stratification, **dust feedback** (dust “weighs down” the fluid)



Vertical Shear Instability

Exploring the VSI's operational range

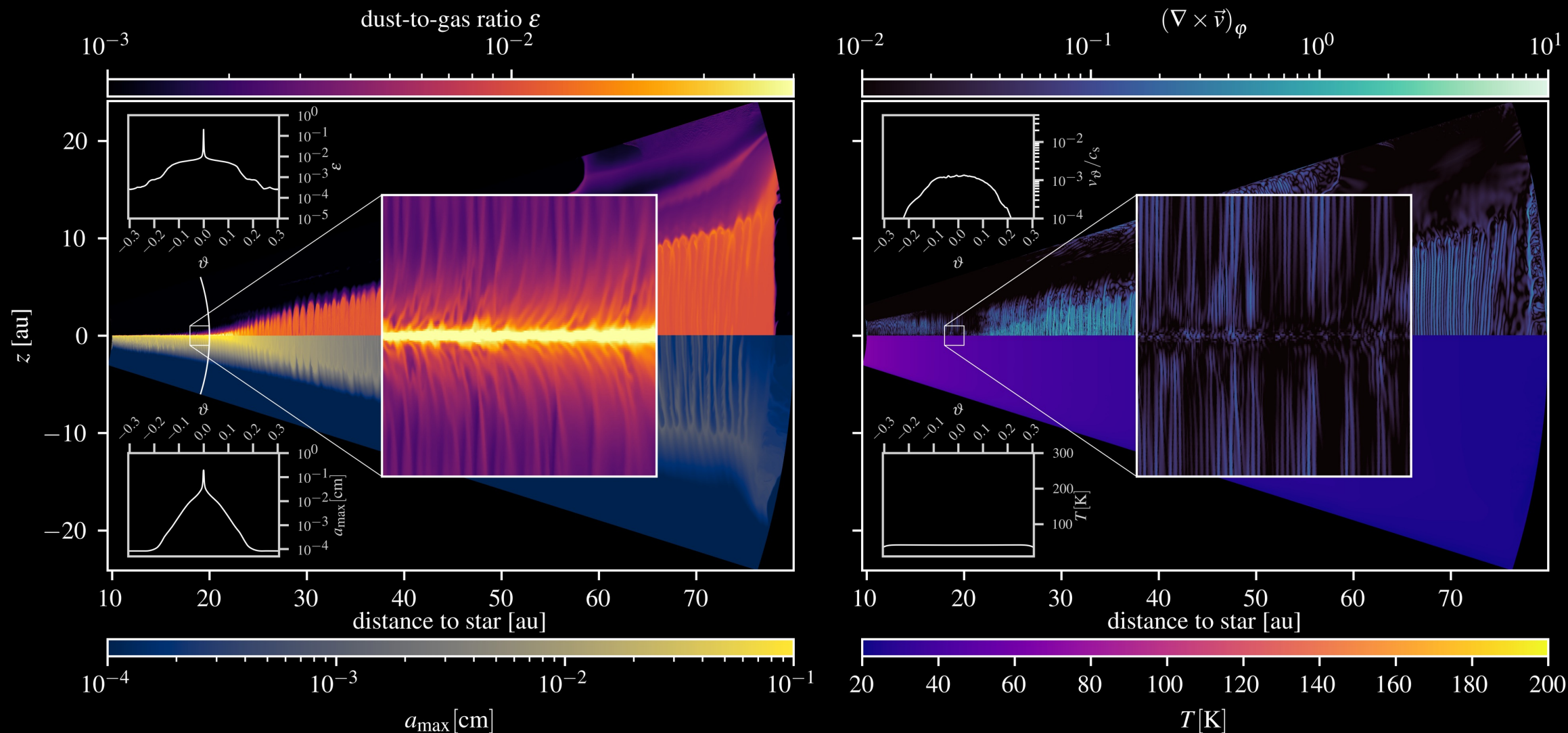
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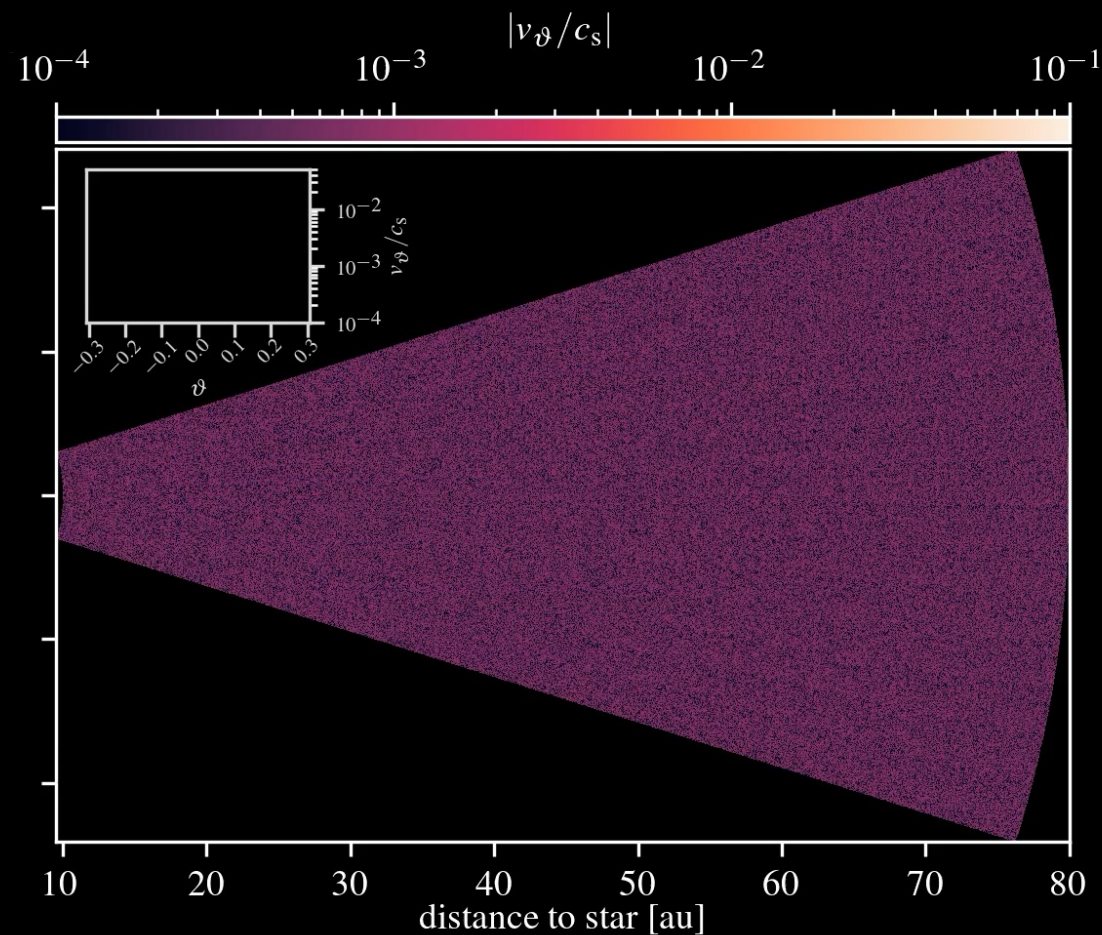
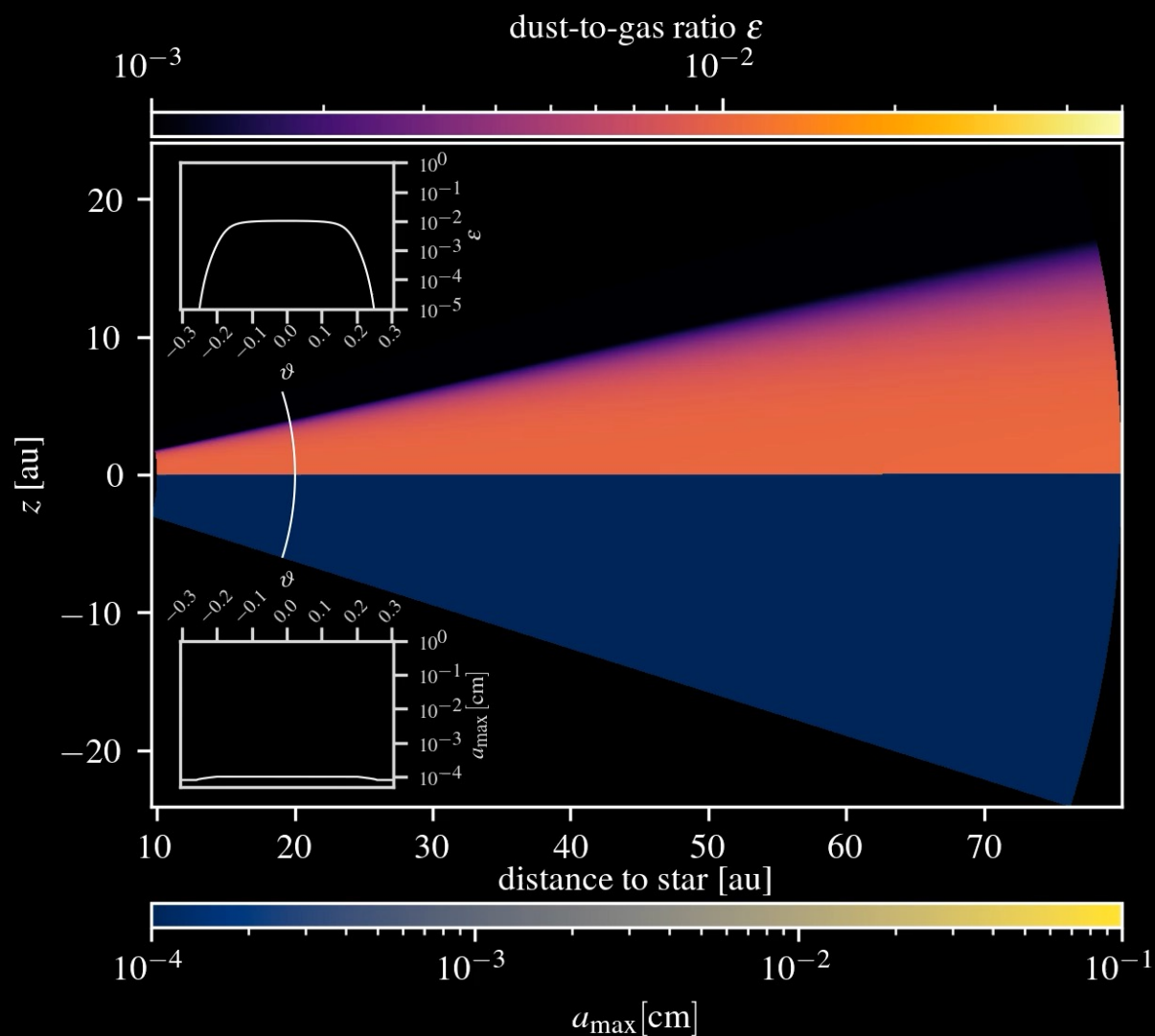
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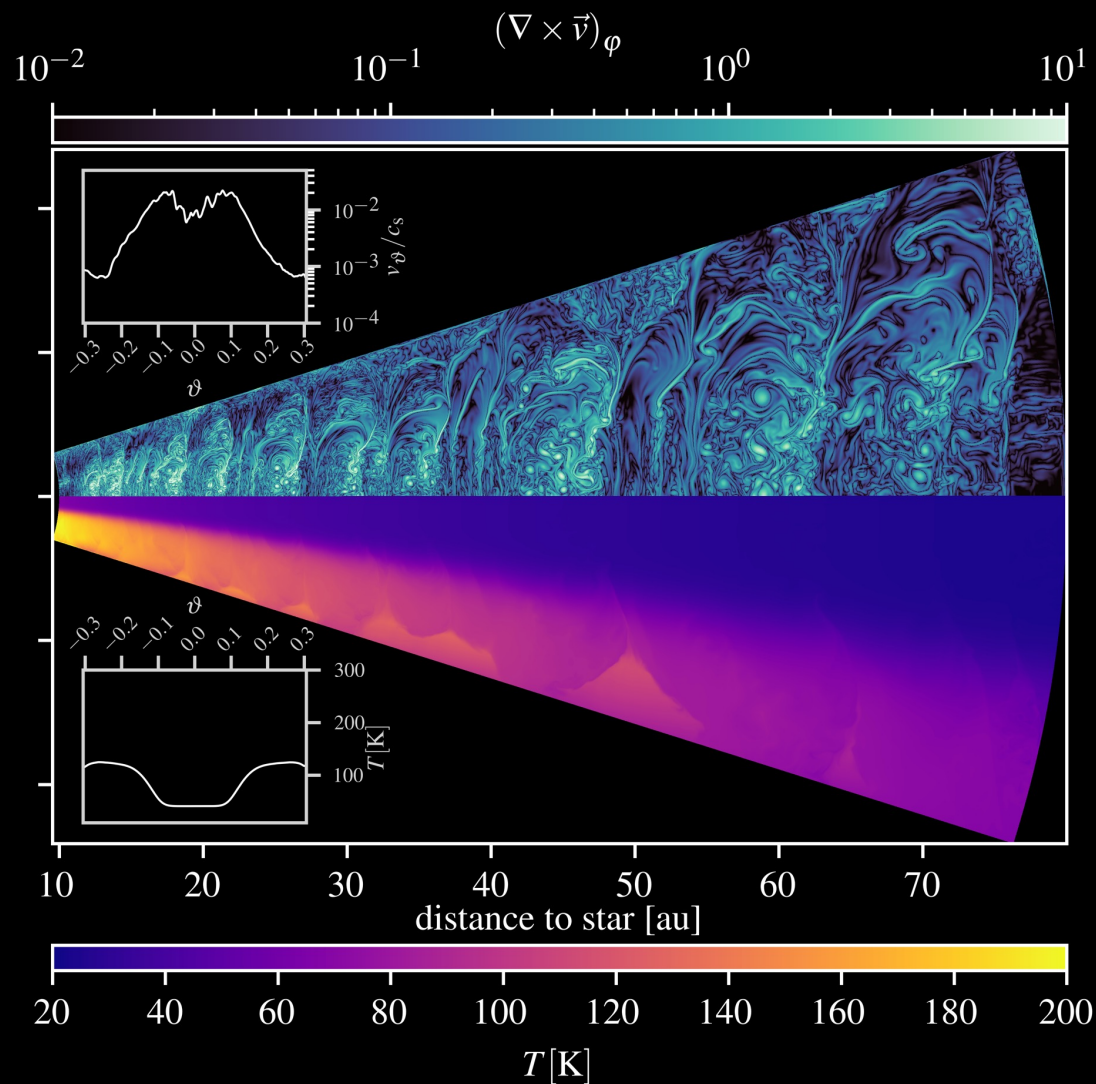
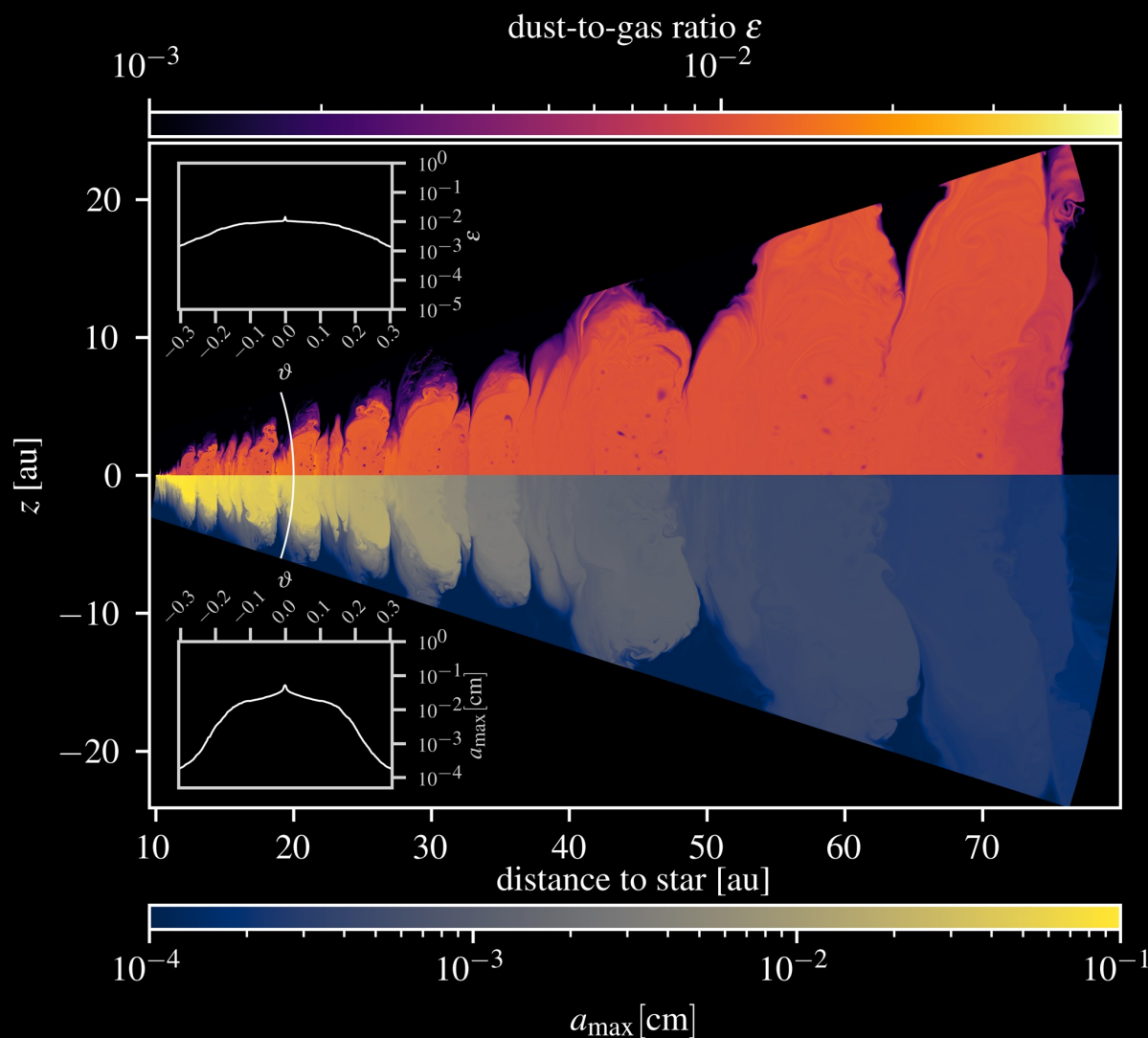
Time Series: VSI as an intermittent state



Vertical Shear Instability

Exploring the VSI's operational range

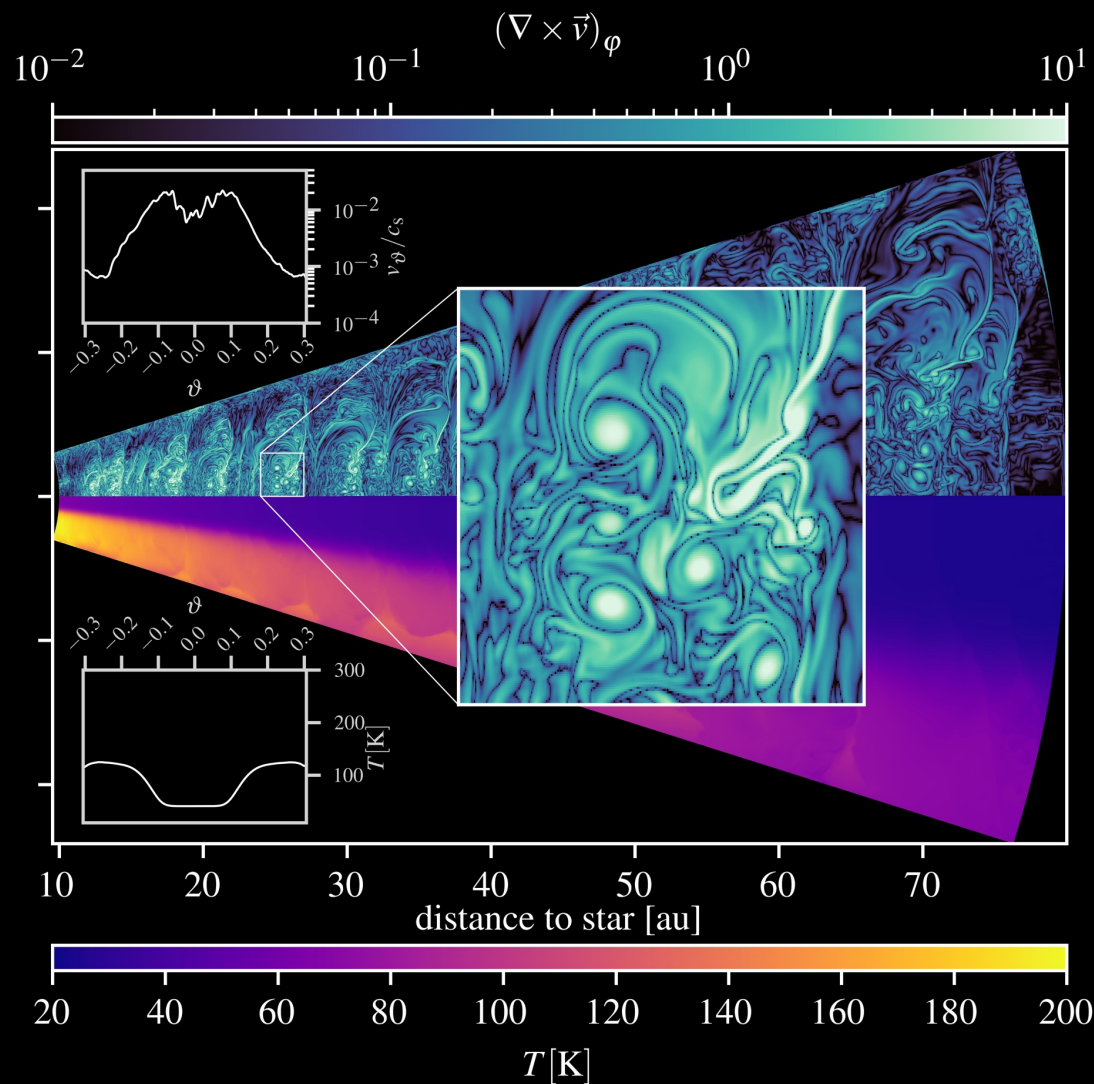
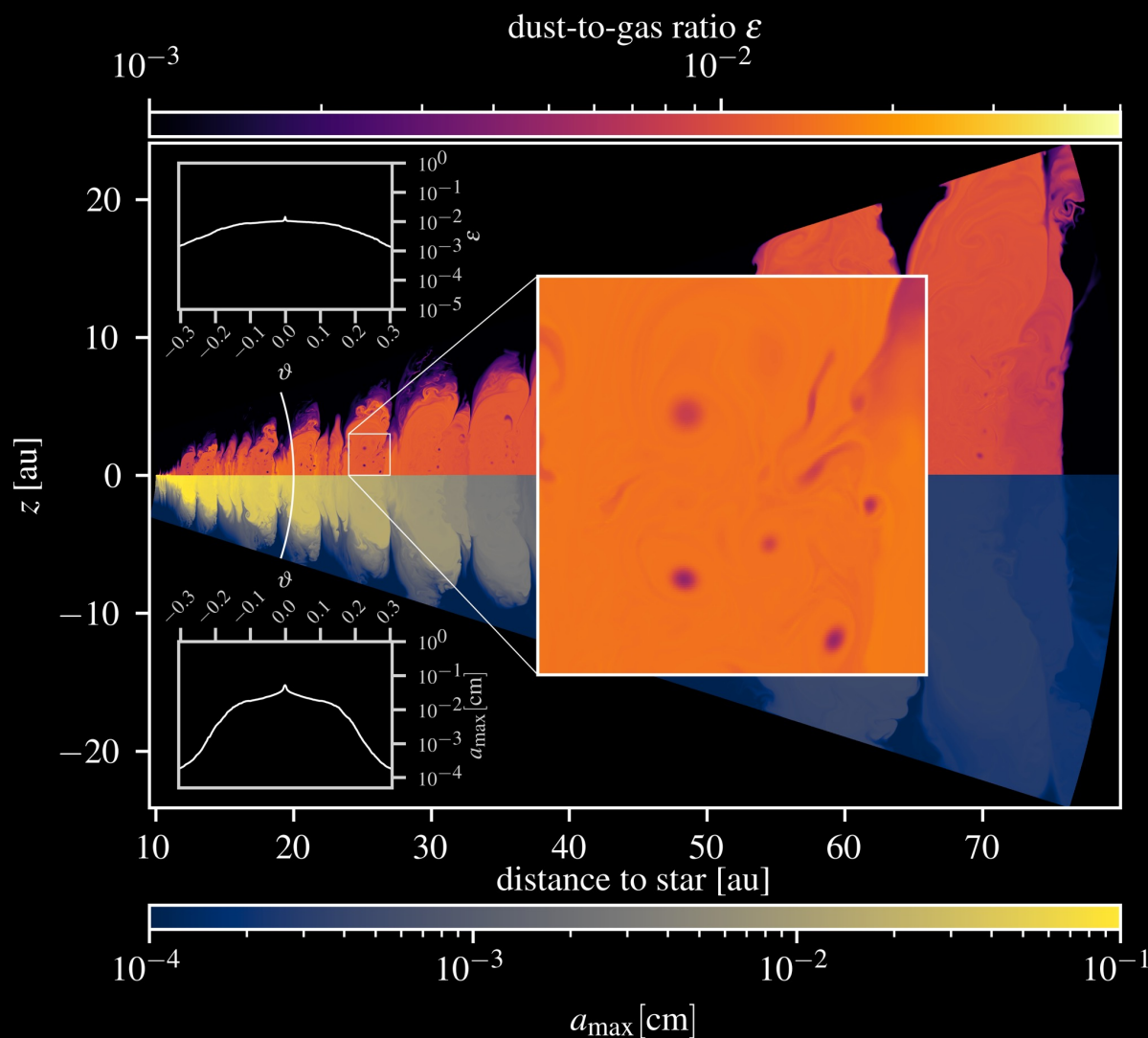
Dust-Gas cooling (non-isothermal), cool midplane and hot atmosphere, dust feedback (dust “weighs down” the fluid)



Vertical Shear Instability

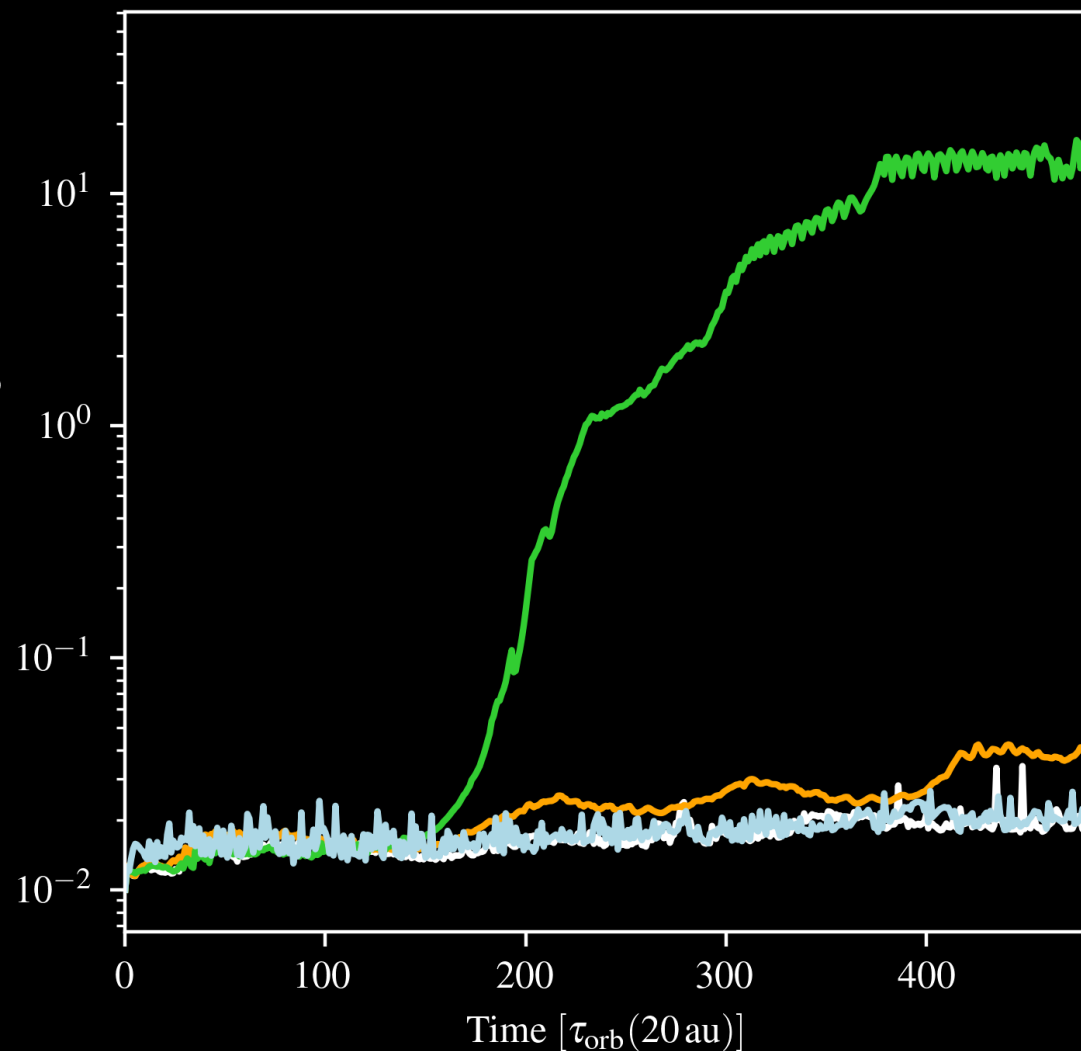
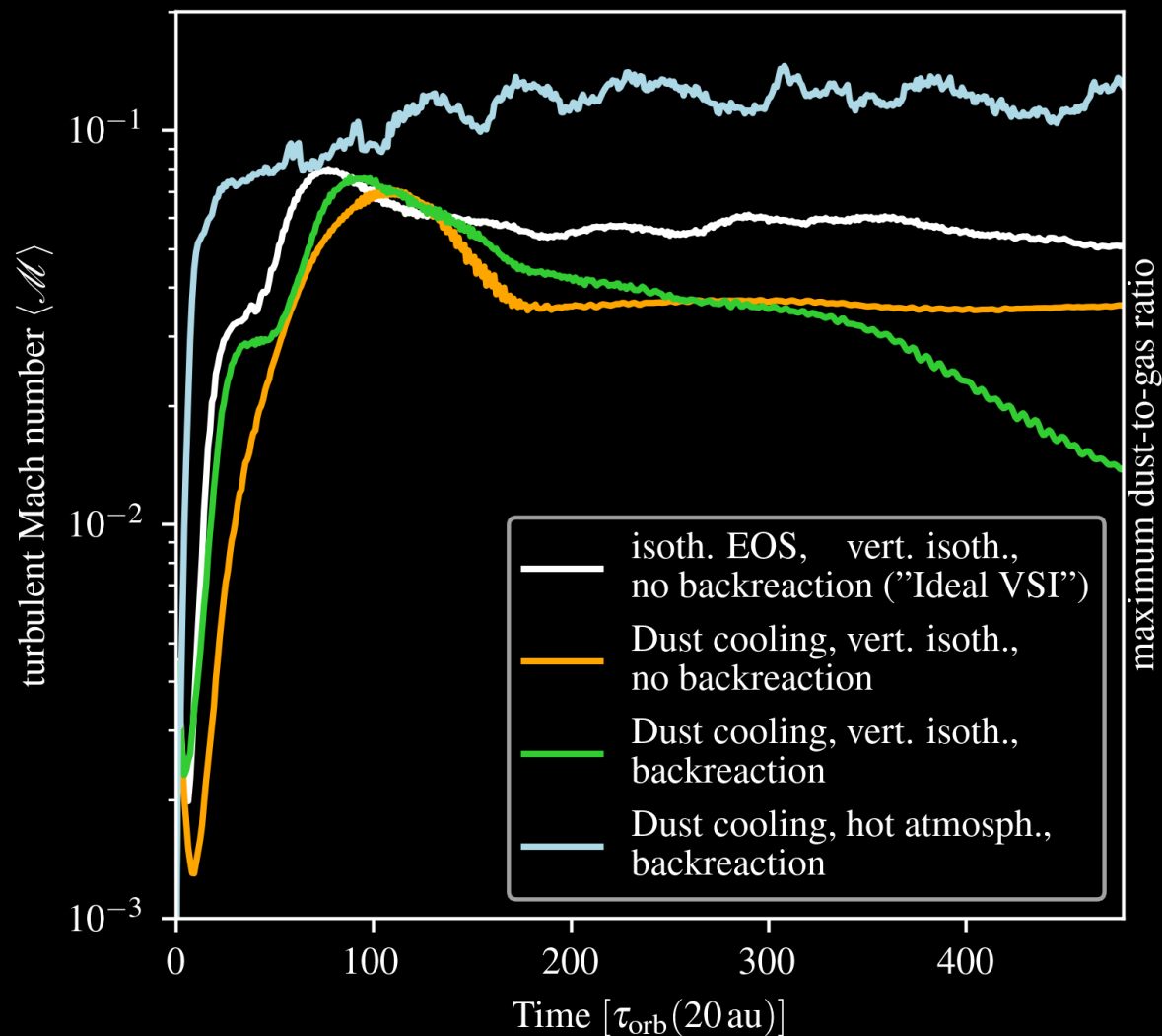
Exploring the VSI's operational range

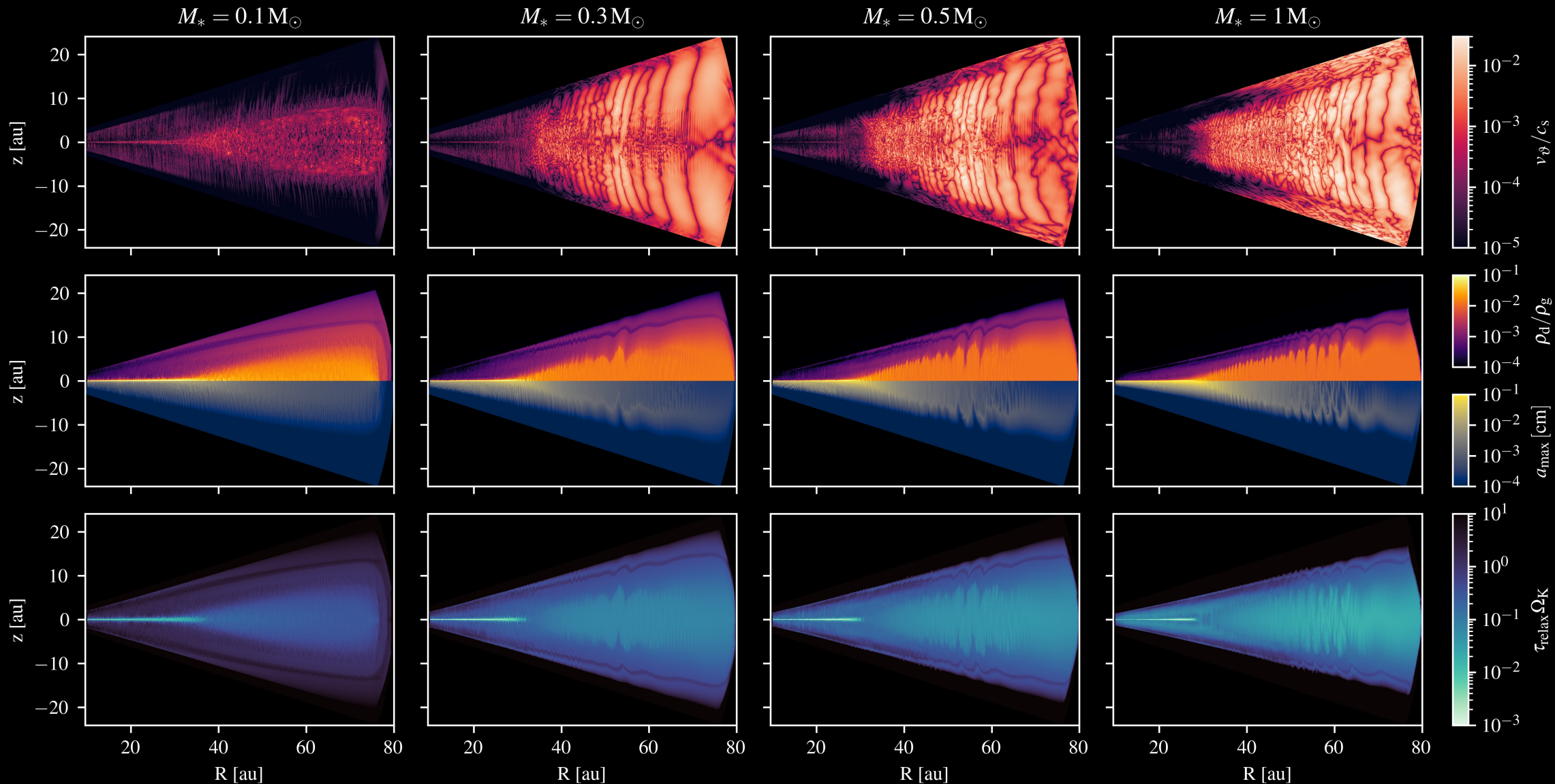
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Vertical Shear Instability

Turbulent Mach Numbers and Dust Sedimentation





- The VSI is highly dependent on the dust physics of the model:
Dust-gas cooling, backreaction, and coagulation determine whether the VSI can drive turbulence in protoplanetary disks.

VSI might just be an intermittent state in some disks, depending on the dust properties (e.g., v_{frag} , ρ_m , etc.)

- Fully coupling VSI and dust evolution will require further efforts:
 - **Dust temperatures and thermodynamics**
 - **Grain collision velocities based on the actual turbulence in the simulation**
→ **Effect of the VSI on dust coagulation?**

